

World Trade: An Overview

ECON 308 International Economic Relations | Chapter 2

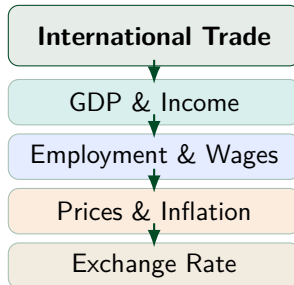
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Why Trade Matters: Canada's Economy Depends on It

- Canada is among the world's **most trade-dependent** advanced economies.
- **Exports $\approx$ 32% of GDP**; imports $\approx$ 33% (2023).
- A 10% drop in Canadian exports is a $\approx$ **3% shock to nominal GDP** - larger than most recessions.
- Trade drives jobs, wages, prices, and the business cycle.



Canada's Top Exports, 2023

Crude oil \$136B · Cars/parts \$65B · Gold \$22B · Natural gas \$17B · Lumber \$14B. **75% of all Canadian exports go to the United States.**

The Trade–Macro Nexus: GDP and Net Exports

In the expenditure approach to GDP:

$$Y = C + I + G + \underbrace{(X - M)}_{\text{Net exports (NX)}}$$

Y = real GDP

I = business investment

X = exports

C = household consumption

G = government spending

M = imports

$NX = X - M$ = **trade balance**

$NX > 0$: trade surplus (Canada exports more than it imports)

$NX < 0$: trade deficit

Canada's $(X-M)/GDP$

Year	X/Y	$(X - M)/Y$
2000	45%	+5.0%
2008	33%	+0.5%
2015	32%	-3.2%
2022	32%	+1.1%

Canada's trade balance swings between surplus and deficit with commodity prices and US demand - both *international* forces with large *domestic* consequences.

Key insight: When exports fall (e.g. US recession), NX falls, and so does Y – directly contracting the Canadian economy.

Learning Objectives

After this lecture you will be able to:

1. Describe why large, nearby economies trade the most with each other.
2. State and interpret the **gravity model** at three levels: intuition, equation, and general equilibrium.
3. Solve a **numerical gravity problem** and interpret the coefficients.
4. Explain why national **borders** and **distance** substantially deter trade even in the digital era.
5. Trace the **two waves of globalisation** and the forces that reversed each.
6. Characterise how the **composition** of world trade has evolved since 1945.
7. Explain how a trade shock transmits to **Canadian macro outcomes** (GDP, employment, monetary policy).

Reference

Krugman, Obstfeld & Melitz, *International Economics: Theory and Policy*, 12th ed., Chapter 2.

Who Trades with Whom? The Basic Facts

- More than **30% of world output** crosses national borders every year.
- World merchandise trade exceeded **\$25 trillion in 2019**.
- Trade is **highly concentrated**: the top-15 US partners account for **75%** of all US trade.
- Top-5 US partners (2019): **Mexico, Canada, China, Japan, Germany**.
- The pattern: **large** economies and **nearby** economies dominate.

Key concepts

Exports (X): goods/services sold abroad.

Imports (M): goods/services bought from abroad.

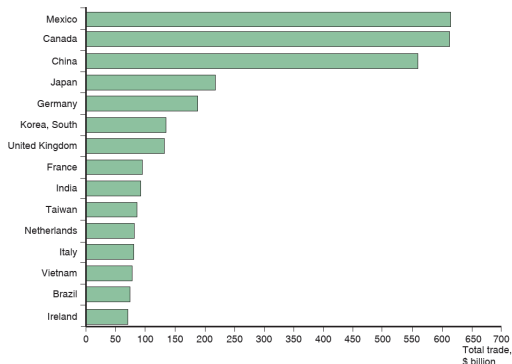
Trade openness = $(X + M)/Y$. Canada: $\approx 65\%$ – very open.

Trade balance = $X - M$. Surplus $\Rightarrow$ net exporter.

Canada–US Trade

Canada–US bilateral trade exceeded **\$700 billion** in 2019 – more than the US traded with the entire EU27 combined.

Figure 2-1: Total US Trade with Major Partners, 2019



Reading the figure:

- Each bar = bilateral trade (exports + imports).
- Top-15 partners = **75%** of all US trade.
- **Mexico and Canada dominate:** proximity + USMCA.

Puzzle

Italy (\$2.0T GDP) is larger than Canada (\$1.9T), yet Canada trades *far* more with the US. Why? Size alone can't explain it.

Size Matters: GDP Predicts Trade Shares

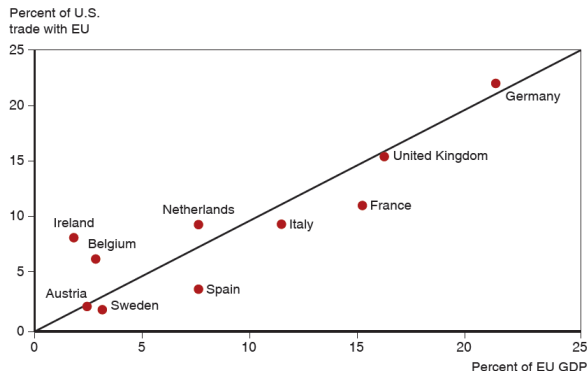


Figure 2-2: Each European country's share of US–Europe trade vs. its share of European GDP.

- Points cluster near the **45° line**: GDP share $\approx$ trade share.
- Bigger economy $\Rightarrow$ more to sell and more to buy $\Rightarrow$ more trade.

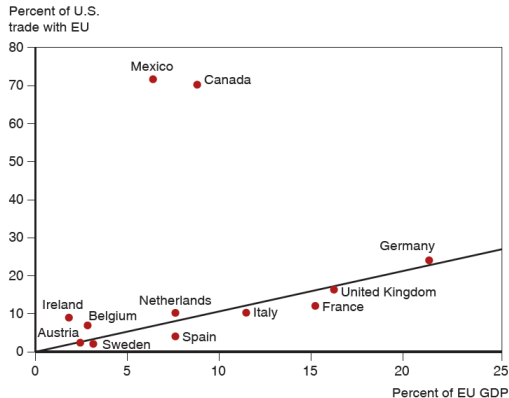
Outliers Tell a Story

Ireland: US tech MNCs host EU HQs there; shared language.

Netherlands: Rotterdam re-exports inflate bi-lateral data.

Outliers reveal what GDP alone misses: culture, institutions, geography.

Figure 2-3: Trade Agreements Boost Trade Beyond GDP



Adding Canada and Mexico to the comparison:

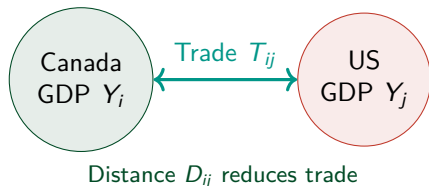
- **Both are far above the line** — they trade much more than GDP alone predicts.
- This excess trade represents the **NAFTA/USMCA effect**: zero tariffs plus deep supply-chain integration.

The Gravity Model of Trade – Intuition

Core Idea

Bilateral trade between two countries is:

- **Proportional to economic size (GDP):**
bigger economies produce more and consume more imports.
- **Inversely proportional to distance:**
farther apart $\Rightarrow$ higher transport cost, weaker information, less trust $\Rightarrow$ less trade.



Canada's Lucky Geography

Canada sits beside the world's largest economy (\$27T GDP) at minimal effective distance. The gravity model predicts enormous Canada–US trade – and that is exactly what we observe.

The Gravity Model – The Equation

The Gravity Equation

$$T_{ij} = A \cdot \frac{Y_i^\alpha Y_j^\beta}{D_{ij}^\gamma} \cdot \exp(\mathbf{z}'_{ij} \delta)$$

T_{ij} : bilateral trade from country i to j

Y_i, Y_j : GDP of exporter and importer

D_{ij} : bilateral distance (great-circle km)

$\mathbf{z}_{ij}$: frictions (language, border, FTA, colonial tie)

A : scale constant

$\alpha, \beta \approx 1$: GDP elasticities

$\gamma \approx 0.7\text{--}1.0$: distance elasticity

δ : friction coefficients

Log-linear form (used in OLS):

$$\ln T_{ij} = c + \alpha \ln Y_i + \beta \ln Y_j - \gamma \ln D_{ij} + \mathbf{z}'_{ij} \delta$$

Bilateral feature $\mathbf{z}_{ij}$	Effect on T
Common language	+35–55%
Contiguous border	+25–50%
FTA membership	+40–80%
Colonial relationship	+20–40%
Currency union	+70–120%

Source: Head & Mayer (2014)

Reading Gravity Coefficients: What Do They Tell Us?

Log-linear gravity:

$$\ln T = c + \underbrace{\alpha \ln Y_i}_{\text{exporter size}} + \underbrace{\beta \ln Y_j}_{\text{importer size}} - \underbrace{\gamma \ln D_{ij}}_{\text{distance}}$$

Interpretation:

- $\alpha = \beta = 1$: 10% bigger economy $\Rightarrow$ 10% more trade. Proportional to size.
- $\gamma = 1$: 10% farther $\Rightarrow$ 10% *less* trade. Robust across all samples since 1970.
- **FTA dummy** $\delta \approx 0.5$: $e^{0.5} \approx 1.65$, so FTA raises trade $\approx 65\%$ (holding GDP and distance fixed).

Gravity Model – Worked Example: Setup

Worked Example

$$T_{ij} = A \cdot \frac{Y_i Y_j}{D_{ij}}, \quad A = 1 \text{ (\$billion, km)}.$$

Data:

Country pair	GDP product	Distance
A (Canada) – B (USA)	\$2,000B × \$15,000B	1,000 km
A (Canada) – C (Japan)	\$2,000B × \$5,000B	10,000 km

Questions:

- (a) Compute T_{AB} and T_{AC} .
- (b) Find the ratio T_{AB}/T_{AC} .
- (c) A CPTPP-style agreement **halves** effective distance to Japan. Compute new T_{AC}^{FTA} .
- (d) What does “effective distance” capture beyond physical km?

Try it now – solution on next slide.

Gravity Model – Worked Example: Solution

(a) Predicted trade:

$$T_{AB} = \frac{2,000 \times 15,000}{1,000} = \mathbf{30,000}$$

$$T_{AC} = \frac{2,000 \times 5,000}{10,000} = \mathbf{1,000}$$

(b) Ratio: $T_{AB}/T_{AC} = \mathbf{30}$

Canada trades **30× more** with the US than with Japan.

(c) CPTPP halves distance to 5,000 km:

$$T_{AC}^{\text{FTA}} = \frac{2,000 \times 5,000}{5,000} = \mathbf{2,000} \quad (\text{doubled})$$

(d) “Effective distance” captures:

- **Tariffs:** $t\%$ tariff $\Rightarrow$ higher effective price.
- **Regulatory barriers:** standards, certificates, inspections.
- **Information costs:** unfamiliarity with distant markets.
- **Language/legal:** harder to negotiate and enforce contracts.

Why Distance Still Matters in the Digital Age

Empirical finding

A **1% increase in distance** reduces bilateral trade by **0.7–1.0%** even controlling for GDP. This effect has *not* declined since 1970.

Five channels:

1. **Transport costs:** bulk goods (oil, grain, lumber, cars) are expensive to ship far.
2. **Time costs:** just-in-time supply chains demand proximity.
3. **Information and trust:** close partners are better known.
4. **After-sale service:** technical support requires local presence.
5. **Regional networks:** USMCA, EU, ASEAN amplify neighbour trade.

BC Lumber: Distance vs. Demand

BC exports $\approx 80\%$ of softwood lumber to the US despite growing Asian demand.

Why? Ocean freight to China $\approx 4\times$ US corridor cost. Distance costs dominate even in the digital era for **physical goods**.

The Home Bias Puzzle: Borders Deter More Than We Think

McCallum (1995) – a landmark finding:

Comparing Canadian interprovincial trade with Canada–US trade at the same distance, McCallum found that:

Canadian provinces traded $\approx 20\times$ more with each other than with equally distant US states.

This finding shocked economists – CUSFTA had *already* eliminated most tariffs. The border effect persisted.

Revised estimates (post-NAFTA) put it at $\approx 10\text{--}12\times$, still enormous.

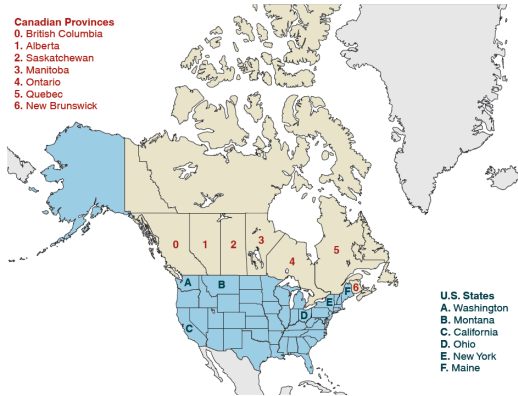
Interpreting the Border Effect

The Canada–US border acts as if it adds $\approx 75,000$ km of extra distance (Anderson & van Wincoop 2003).

This huge implicit barrier is driven by:

- Different regulatory standards and product certifications
- Currency exchange costs and USD/CAD uncertainty
- Different legal systems
- Customs documentation even under USMCA
- Cultural and information differences

Figure 2-4: BC Trade with Canadian Provinces vs. US States



Reading Figure 2-4:

- Plots BC trade as % of GDP against distance from Vancouver.
- **Red (Canadian provinces):** trade is much higher at every distance.
- **Blue (US states):** trade is much lower, even for close states.
- The **gap** between the lines = the *border effect*.

Table 2-1: BC Trade as % of GDP (2009)

Partner	GDP (C\$B)	BC trade/GDP
<i>Canadian Provinces</i>		
Ontario	602	6.9%
Alberta	323	2.6%
Quebec	297	1.0%
<i>US States</i>		
Washington	350	0.4%
California	1,750	0.3%
New York	1,140	0.1%

Source: Statistics Canada; US Dept. of Commerce

Key observations:

- Ontario (6.9%) vs. Washington State (0.4%) at similar distances – a $17\times$ gap.
- California is $3\times$ larger than Ontario but trades far less with BC.
- Border costs explain the entire gap: currency, customs, standards, legal systems.

Behind-the-Border Barriers: Non-Tariff Costs

Even under zero-tariff CUSMA, barriers remain:

1. **Regulatory differences:** safety, environmental, labelling standards.
2. **Rules of origin:** 75% North American content for autos; creates compliance costs.
3. **Currency risk:** CAD/USD fluctuations add contract uncertainty.
4. **Legal differences:** contract enforcement
5. **“Buy American”:** US government procurement preferences.

Frontier: cutting non-tariff barriers matters more than further tariff cuts.

Two Waves of Globalisation: The First Wave (1870–1913)

What drove the first wave?

- **Technology:** steamships cut ocean freight costs by 70%; railways opened continental interiors; telegraph enabled real-time price information.
- **Monetary framework:** classical gold standard fixed exchange rates and enabled large cross-border capital flows.

Result: World trade/GDP rose from $\approx 5\%$ (1850) to $\approx 12\%$ (1913).

Retreat (1914–1945) and the Second Wave (1945–)

Interwar collapse:

- WWI shattered trade finance and supply chains.
- Great Depression + Smoot-Hawley (1930): US tariff $\approx 45\%$.
- Retaliatory tariffs worldwide: world trade **fell 66%** in 1929–33.

Second wave (1945–):

- GATT (1947); Bretton Woods; containerisation (1956).
- Trade/GDP exceeded pre-WWI peak by 1970.
- WTO (1995); China WTO (2001); GVC explosion.
- Post-2008 slowdown; post-2020 fragmentation.

The Political Lesson

Globalisation requires *both* technology *and* political commitment.

Smoot-Hawley reversed decades of liberalisation in two years.

GATT/WTO is the institutional safeguard – but faces strain: US-China tech war, WTO appellate body crisis (2019).

Figure 2-5: The Rise, Fall, and Rise of World Trade



Reading Figure 2-5:

- **First peak** ≈ 1913 : first globalisation wave.
- **Deep trough** 1914–45: wars, Depression, Smoot-Hawley.
- **Recovery** 1950s–70s: GATT, containerisation.
- **New peak** 1990s–2008: WTO, China, GVCs.
- **Post-2008**: trade grew slower than GDP.

Canadian Example

Canada's export/GDP closely mirrors the world. 2000s boom: NAFTA + China commodity demand.

Post-2020: The Fragmentation of Global Trade

New forces reshaping global trade since 2020:

- **COVID-19:** supply chain disruptions revealed concentration risks (semiconductors, PPE, pharmaceuticals).
- **US-China tech war:** export controls on chips, AI, telecom.
- **Russia-Ukraine war:** energy and food commodity markets disrupted.
- **“Friend-shoring”:** shifting supply chains to geopolitically aligned partners.
- **Industrial policy resurgence:** US Inflation Reduction Act, EU Green Deal — subsidies for domestic production.

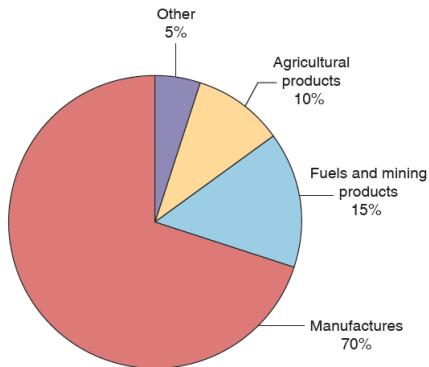
Canada's Opportunity and Risk

Opportunity: As a trusted US ally, Canada benefits from friend-shoring. EV battery plants (Volkswagen, Stellantis in Ontario) are a direct result.

Risk: If global GVCs fragment, Canada's export diversity could fall further toward US dependence.

Canadian trade policy under fragmentation must balance **CUSMA integration** with **CPTPP** and **CETA diversification**.

What Do Countries Trade? Composition of World Trade



World merchandise trade (2017):

- **Manufactures** $\approx 68\%$: machinery, cars, electronics, chemicals.
- **Fuels/mining** $\approx 15\%$: oil, gas, coal, metals.
- **Agriculture** $\approx 10\%$: wheat, soybeans, coffee, fish.

Canada's Unusual Trade Basket

Canada exports *all three* categories: oil (energy), cars/aircraft (manufacturing), wheat/canola/potash (agriculture).

This dual structure makes Canada sensitive to both commodity-price shocks *and* manufacturing-demand shocks – unusual among G7 peers.

Table 2-2: The Rise of Manufacturing Trade

Country	1910 exp.	1910 imp.	2002 exp.	2002 imp.
UK	75%	24%	82%	79%
US	47%	40%	84%	77%

Source: Kuznets (1967); WTO

The shift:

- **1910:** UK exported manufactures, imported primary goods. *Inter-industry* trade.
- **2002:** Both UK exports *and* imports are manufactures. This is *intra-industry* trade.

Inter-industry vs. intra-industry:

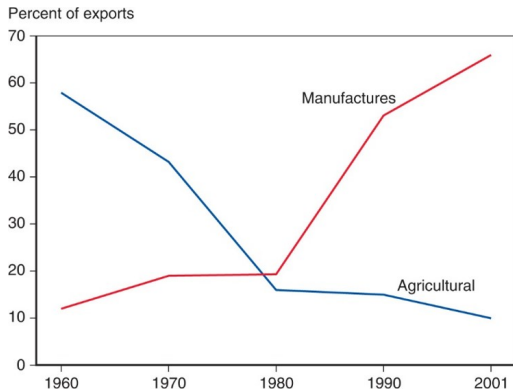
Inter-industry: Canada exports wheat, imports electronics. Explained by comparative advantage (Ch. 3–5).

Intra-industry: Canada both exports and imports cars. Explained by economies of scale (Ch. 7).

Macro Connection

Manufacturing demand is more cyclical than agriculture. Countries specialised in manufactures have more volatile trade balances – relevant for Canadian monetary policy.

The Transformation of Developing Country Exports



A remarkable structural shift:

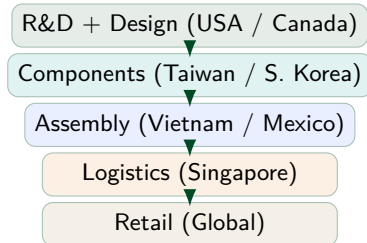
- **1960:** $\approx 58\%$ of developing exports were primary commodities.
- **2001:** $\approx 65\%$ were manufactures.
- **China:** $>90\%$ manufactures; moving toward AI and semiconductors.

Country	Mfg.% 1960	Mfg.% 2019
China	10	93
South Korea	20	89
Bangladesh	5	90

Global Value Chains: “Made in the World”

Modern production is internationally fragmented:

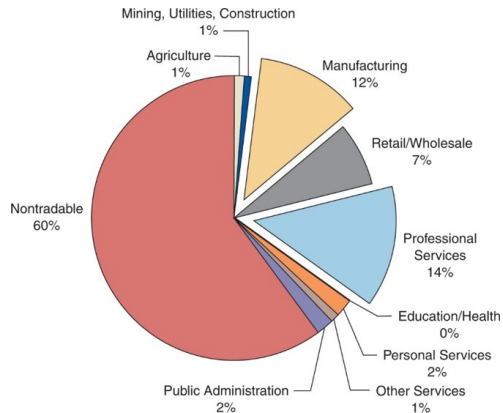
- Design, components, assembly, and retail done in *different countries*.
- A \$500 smartphone may cross **6 borders** before reaching a consumer.
- Gross exports *overstate* domestic content:
Chinese re-exports of Malaysian components count as Chinese exports.



Policy Issue

Tariffs on intermediate inputs hurt domestic assemblers.
US 2018 steel tariffs raised costs for US automakers – including USMCA Canadian plants.

Service Trade: The New Frontier



Services trade is fast-growing:

- World services trade $\approx$ \$6T (2019): fastest-growing category.
- **Tradable:** IT, finance, engineering, education, radiology.
- **Non-tradable:** haircuts, local retail, most healthcare.

Canadian Example

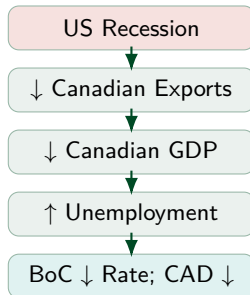
Canada: **net importer** of IT services (US/India), **net exporter** of financial services, engineering consulting, education.

$\approx$ 850,000 international students/year – a large “invisible” Canadian export.

When the US Sneezes, Canada Catches a Cold

Trade transmission mechanism:

1. **US recession:** US aggregate demand falls.
2. **Canadian exports fall:** $\Delta X < 0 \Rightarrow \Delta NX < 0$.
3. **Canadian GDP falls:** $\Delta Y < 0$.
4. **Unemployment rises:** Ontario auto, BC forestry, Alberta oil.
5. **Bank of Canada responds:** cuts overnight rate.
6. **CAD depreciates:** partial export offset.



2008–09 Case

Canada entered recession months after the US financial crisis despite no domestic housing bubble. BoC cut rates from 4.5% to 0.25% within one year.

Commodity Prices and Canada's Terms of Trade

Terms of trade (ToT) = $P_{\text{exports}}/P_{\text{imports}}$.

For Canada (resource exporter), oil prices drive ToT:

- Oil $\uparrow \Rightarrow$ ToT improve $\Rightarrow$ real income rises above real GDP.
- Oil $\downarrow \Rightarrow$ ToT worsen $\Rightarrow$ Alberta/Saskatchewan recession.

Period	Oil price	Canadian effect
2002–14	\$25→\$110	ToT $\uparrow$ 40%; boom
2014–16	\$110→\$45	Alberta recession; CAD \$0.68
2021–22	\$45→\$110	ToT recovered; CAD strengthened

Dutch Disease: When Resource Wealth Hurts Manufacturing

The Dutch Disease mechanism:

1. Commodity boom raises Canada's export revenues.
2. Increased foreign demand for CAD pushes CAD higher.
3. Higher CAD makes Canadian manufactured exports more expensive for foreign buyers.
4. Manufacturing exports fall; factory closures and job losses in Ontario/Quebec.
5. Resource sector booms while manufacturing sector "hollows out."

Historical case:

2007–2012: CAD reached parity with USD. Ontario auto and forestry exports fell sharply. "Dutch Disease" debate dominated Canadian economic policy discussion.

Named After the Netherlands

In the 1960s, the Netherlands discovered North Sea natural gas. Gas export revenues pushed up the Dutch guilder.

Dutch manufactured exports became uncompetitive. Manufacturing employment fell sharply – even though the country became richer overall.

The **distributional effect**: resource-sector workers gain; manufacturing-sector workers lose – connecting to the Specific Factors Model (Ch. 4).

Exchange Rates and Trade Competitiveness: A Preview

Trade and exchange rates are deeply linked:

- Let e = USD per 1 CAD ($\uparrow e$ = CAD appreciates).
- Price of Canadian export in USD:
 $P_{\text{USD}} = e \cdot P_{\text{CAD}}$.
- **CAD appreciates** ($e \uparrow$): Canadian exports more expensive $\Rightarrow$ volumes fall.
- **CAD depreciates** ($e \downarrow$): Canadian exports cheaper $\Rightarrow$ volumes rise.

2007–2012: CAD at parity with USD. Canadian manufacturers lost competitiveness; BC lumber exports fell sharply. The “Dutch Disease” debate.

Real exchange rate:

$$q = \frac{e \cdot P^*}{P}$$

P^* = US price level, P = Canadian

$q \uparrow$ = Canada more competitive

$q \downarrow$ = Canada less competitive

Full treatment in macroeconomics. Here: the direction of the effect matters for understanding trade volume.

Should Canada Diversify Away from the US?

The case for diversification:

- 75% US dependence creates macroeconomic vulnerability.
- US political risk: tariff threats, CUSMA renegotiations.
- CPTPP (2018) and CETA (2017) open new corridors.

The gravity model's answer:

- US proximity plus USMCA will remain dominant.
- Effective distance to Japan ($\approx 10,000$ km) vs. the US ($\approx 1,000$ km) means gravity strongly favours the US.
- Diversification is possible but the model predicts limited impact.

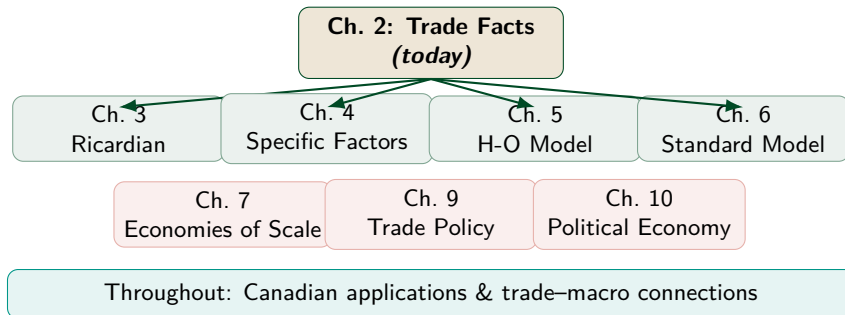
Discussion

1. Can CPTPP realistically cut Canada's US export share below 65%?
2. If the US imposed a 10% tariff on Canadian goods, what would happen to Canadian GDP, employment, and the exchange rate?
3. Is trade diversification genuine policy or political signalling?

Five Key Facts About World Trade

1. **Gravity rules:** trade rises with GDP and falls with distance. Borders remain huge barriers even under FTAs.
2. **Globalisation is political:** it reversed dramatically 1914–45. Technology enables; institutions sustain.
3. **Manufactures dominate;** services trade grows fastest. Developing countries shifted from commodities to manufactures.
4. **GVCs define modern trade:** production is internationally fragmented; gross exports overstate domestic content.
5. **Trade drives Canadian macro:** exports $\approx 32\%$ of GDP; US the dominant partner; commodity prices and US cycles transmit through trade.

Course Roadmap: Where We Go From Here



Discussion Questions and Further Reading

For next class:

1. Gravity predicts Canada will always trade mainly with the US. Does this mean CPTPP is a waste? Defend your view.
2. $\gamma \approx 1$ has not fallen despite the internet. Does this mean digital trade is *not* growing? Explain the difference.
3. If Canada imposed a 10% tariff on US goods (breaking CUSMA), what happens to Canadian GDP, inflation, and the exchange rate?

Key papers:

- McCallum (1995), "National Borders Matter," *AER*.
- Anderson & van Wincoop (2003), "Gravity with Gravitas," *AER*.
- Head & Mayer (2014), "Gravity Equations: Workhorse, Toolkit, Cookbook."

Key Takeaways

- Gravity: $GDP \uparrow \Rightarrow \text{trade} \uparrow$; $\text{distance} \uparrow \Rightarrow \text{trade} \downarrow$.
- Borders: huge implicit barriers even under FTAs.
- Globalisation: comes in waves; institutions sustain it.
- Composition: shifting toward services and GVCs.
- For Canada: trade *is* macroeconomic policy.

Next Lecture

Ch. 3 – Ricardian Model: Why do countries trade?
What determines export patterns?